

EN3160 Assignment 01

Intensity Transformations and Neighborhood Filtering

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[GitHub Repository](#)

```
In [1]: import cv2 as cv, numpy as np, matplotlib.pyplot as plt
        from pathlib import Path
        IMAGE_DIR, OUTPUT_DIR = Path("images"), Path("outputs")
        OUTPUT_DIR.mkdir(exist_ok=True)
```

Q1 — Piecewise Intensity Transformation

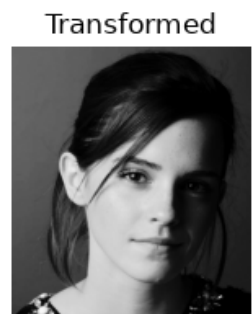
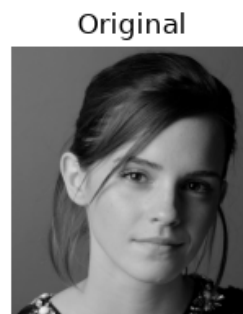
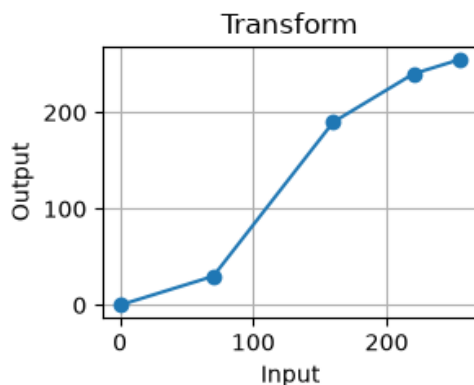
A piecewise-linear intensity transformation was applied using selected breakpoints to expand the mid-intensity range and improve contrast.

```
In [2]: # Q1 - Piecewise intensity transformation

def intensity_transform(im, bp):
    bp=np.asarray(bp,dtype=np.float32); out=np.zeros_like(im,dtype=np.float32)
    for i in range(len(bp)-1):
        x0,y0=bp[i]; x1,y1=bp[i+1]
        m=(im>=x0)&(im<=(x1 if i==len(bp)-2 else x1-1))
        out[m]=y0+(y1-y0)*(im[m]-x0)/(x1-x0)
    return np.clip(out,0,255).astype(np.uint8)

img_q1=cv.imread(str(IMAGE_DIR/"q1_portrait.png"),cv.IMREAD_GRAYSCALE)
bp=np.array([[0,0],[70,30],[160,190],[220,240],[255,255]],dtype=np.float32)
q1_out=intensity_transform(img_q1,bp)

fig,ax=plt.subplots(1,3,figsize=(8,2.6))
ax[0].plot(bp[:,0],bp[:,1],'-o'); ax[0].set_title("Transform"); ax[0].set_xlabel("I")
ax[1].imshow(img_q1,cmap="gray",vmin=0,vmax=255); ax[1].set_title("Original")
ax[2].imshow(q1_out,cmap="gray",vmin=0,vmax=255); ax[2].set_title("Transformed")
for a in ax[1:]: a.axis("off")
plt.tight_layout(); plt.show()
```



Discussion: The transformation increases the contrast of the mid-intensity regions, making facial and hair details more distinct while keeping the output within the valid 8-bit range.

Q2 — White and Gray Matter Enhancement

Two piecewise-linear intensity transformations were designed to emphasize different tissue intensity ranges in the brain proton-density image.

```
In [3]: # Q2 - White and gray matter enhancement

img_q2 = cv.imread(str(IMAGE_DIR/"q2_brain.png"), cv.IMREAD_GRAYSCALE)

white_bp = np.array([
    [0,0], [140,35], [165,70], [195,220], [220,245], [255,255]
], dtype=np.float32)

gray_bp = np.array([
    [0,0], [130,30], [150,70], [175,210], [200,235], [255,255]
], dtype=np.float32)

white_enh = intensity_transform(img_q2, white_bp)
gray_enh = intensity_transform(img_q2, gray_bp)

fig, ax = plt.subplots(2,3,figsize=(9,5))

ax[0,0].plot(white_bp[:,0], white_bp[:,1], '-o')
ax[0,0].set_title("White-matter Transform"); ax[0,0].grid()

ax[0,1].plot(gray_bp[:,0], gray_bp[:,1], '-o')
ax[0,1].set_title("Gray-matter Transform"); ax[0,1].grid()

ax[0,2].axis("off")

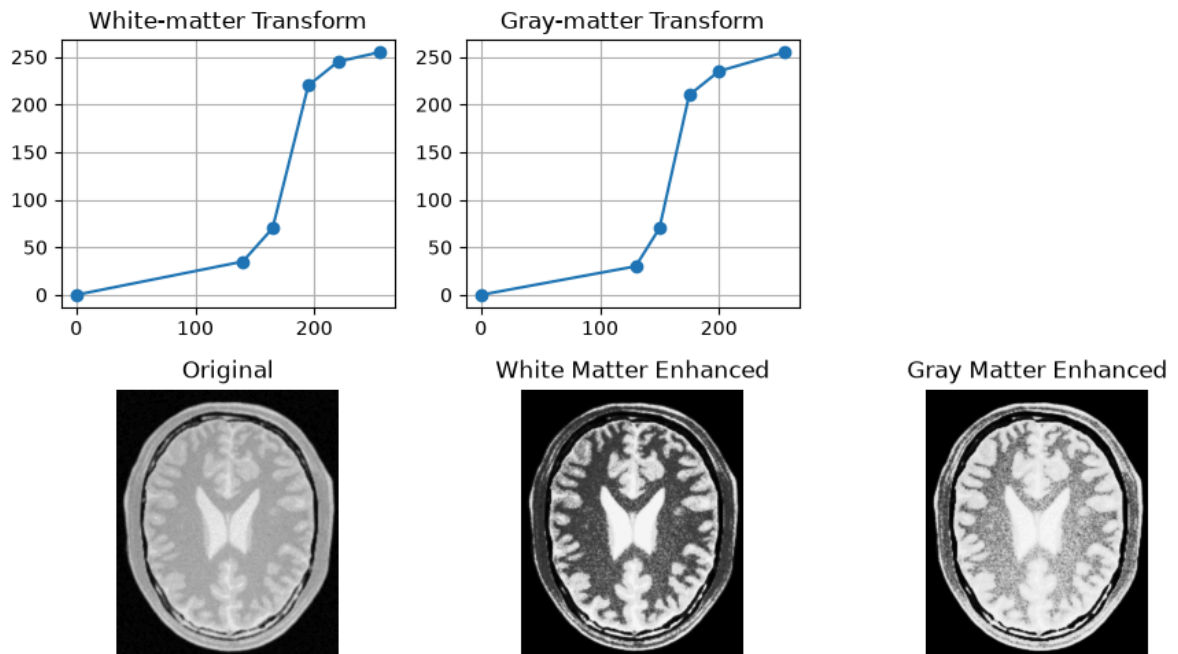
ax[1,0].imshow(img_q2,cmap="gray",vmin=0,vmax=255)
ax[1,0].set_title("Original")

ax[1,1].imshow(white_enh,cmap="gray",vmin=0,vmax=255)
ax[1,1].set_title("White Matter Enhanced")

ax[1,2].imshow(gray_enh,cmap="gray",vmin=0,vmax=255)
ax[1,2].set_title("Gray Matter Enhanced")

for a in ax[1]:
    a.axis("off")

plt.tight_layout()
plt.show()
```



Discussion: The two mappings expand different intensity ranges. The first mapping gives stronger contrast to the selected higher-intensity tissue range, while the second shifts the strongest enhancement toward a slightly lower range. This allows different tissue regions to be emphasized separately rather than simply increasing the overall brightness.

Q3 — Gamma Correction

Gamma correction was applied only to the lightness channel of the image in the Lab color space so that brightness could be adjusted without directly changing the color channels.

```
In [23]: # Q3 - Gamma correction on L channel

img_q3 = cv.imread(str(IMAGE_DIR/"q3_gamma.png"))
lab = cv.cvtColor(img_q3, cv.COLOR_BGR2LAB)
L, a, b = cv.split(lab)

gamma = 0.7
L_new = np.clip(255*(L/255.0)**gamma, 0, 255).astype(np.uint8)

out_q3 = cv.cvtColor(cv.merge([L_new,a,b]), cv.COLOR_LAB2RGB)

fig,ax = plt.subplots(1,3,figsize=(9,3))

ax[0].imshow(cv.cvtColor(img_q3,cv.COLOR_BGR2RGB))
ax[0].set_title("Original")

ax[1].imshow(out_q3)
ax[1].set_title("Gamma Corrected")

ax[2].hist(L.ravel(),bins=256, range=(0,256),alpha=0.5,label="Original")
ax[2].hist(L_new.ravel(),bins=256, range=(0,256),alpha=0.5,label="Corrected")
ax[2].set_title("L-channel Histograms")
ax[2].legend()
```

```

for a1 in ax[:2]:
    a1.axis("off")

plt.tight_layout()
plt.show()

```



Selecting the Gamma Value

Several gamma values were tested before selecting the final value. Values below 1 brighten the image, while values above 1 darken it.

```

In [24]: img_q3 = cv.imread(str(IMAGE_DIR/"q3_gamma.png"))

lab = cv.cvtColor(img_q3, cv.COLOR_BGR2LAB)
L, A_lab, B_lab = cv.split(lab)

gamma_values = [0.5, 0.7, 0.9, 1.2, 1.5]

fig, ax = plt.subplots(2, 3, figsize=(9, 5))

ax[0, 0].imshow(cv.cvtColor(img_q3, cv.COLOR_BGR2RGB))
ax[0, 0].set_title("Original")
ax[0, 0].axis("off")

for i, g in enumerate(gamma_values):
    Lt = np.clip(255*(L/255.0)**g, 0, 255).astype(np.uint8)

    test = cv.cvtColor(
        cv.merge([Lt, A_lab, B_lab]),
        cv.COLOR_LAB2RGB
    )

    ax.ravel()[i+1].imshow(test)
    ax.ravel()[i+1].set_title(f"γ = {g}")
    ax.ravel()[i+1].axis("off")

plt.tight_layout()
plt.show()

```



Discussion: A gamma value of 0.7 was selected because it brightened the dark and mid-level regions while keeping the brighter areas reasonably controlled. The correction was applied only to the L channel, so the a and b color channels were not directly modified.

Discussion: For ($\gamma < 1$), dark and mid-level intensities are increased, while ($\gamma > 1$) makes the image darker. Among the tested values, ($\gamma = 0.7$) gave a useful improvement in the darker rock regions without making the bright parts of the image excessively strong. Therefore, ($\gamma = 0.7$) was selected for the final correction.

Q4 — Vibrance Enhancement

The saturation channel of the HSV image was modified using the given vibrance transformation. Several values of the parameter (a) were tested, and ($a = 0.6$) was selected for the final result.

```
In [5]: # Q4 - Vibrance enhancement

img_q4 = cv.imread(str(IMAGE_DIR/"q4_vibrance.png"))
hsv = cv.cvtColor(img_q4, cv.COLOR_BGR2HSV)
H, S, V = cv.split(hsv)

a, sigma = 0.6, 70
S_new = S.astype(np.float32)
S_new = S_new + a*128*np.exp(-((S_new-128)**2)/(2*sigma**2))
S_new = np.clip(S_new,0,255).astype(np.uint8)

out_q4 = cv.cvtColor(cv.merge([H,S_new,V]), cv.COLOR_HSV2RGB)

x = np.arange(256)
y = np.clip(x + a*128*np.exp(-((x-128)**2)/(2*sigma**2)),0,255)

fig,ax = plt.subplots(1,3,figsize=(9,3))
```

```

ax[0].plot(x,y)
ax[0].set_title("Saturation Transform")
ax[0].set_xlabel("Input S")
ax[0].set_ylabel("Output S")
ax[0].grid()

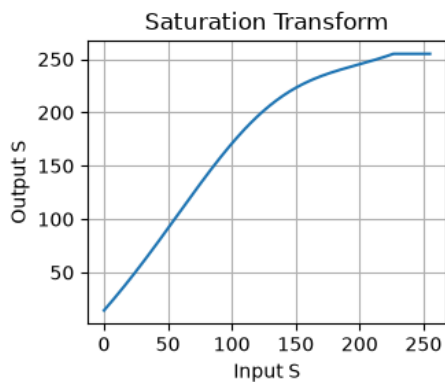
ax[1].imshow(cv.cvtColor(img_q4,cv.COLOR_BGR2RGB))
ax[1].set_title("Original")

ax[2].imshow(out_q4)
ax[2].set_title("Vibrance Enhanced")

for a1 in ax[1:]:
    a1.axis("off")

plt.tight_layout()
plt.show()

```



Implementing the vibrance transformation

```

In [6]: def vibrance_transform(S, a, sigma=70):
        S_float = S.astype(np.float32)

        boost = a * 128.0 * np.exp(
            -((S_float - 128.0) ** 2) / (2.0 * sigma ** 2)
        )

        S_new = S_float + boost

        return np.clip(S_new, 0, 255).astype(np.uint8)

```

```

In [7]: a_values = [0.2, 0.4, 0.6, 0.8, 1.0]

plt.figure(figsize=(15, 7))

for i, a_val in enumerate(a_values):
    S_corr = vibrance_transform(S, a_val)

    hsv_corr = cv.merge([H, S_corr, V])
    bgr_corr = cv.cvtColor(hsv_corr, cv.COLOR_HSV2BGR)
    rgb_corr = cv.cvtColor(bgr_corr, cv.COLOR_BGR2RGB)

    plt.subplot(2, 3, i + 1)

```

```
plt.imshow(rgb_corr)
plt.title(f"a = {a_val}")
plt.axis("off")

plt.tight_layout()
plt.show()
```

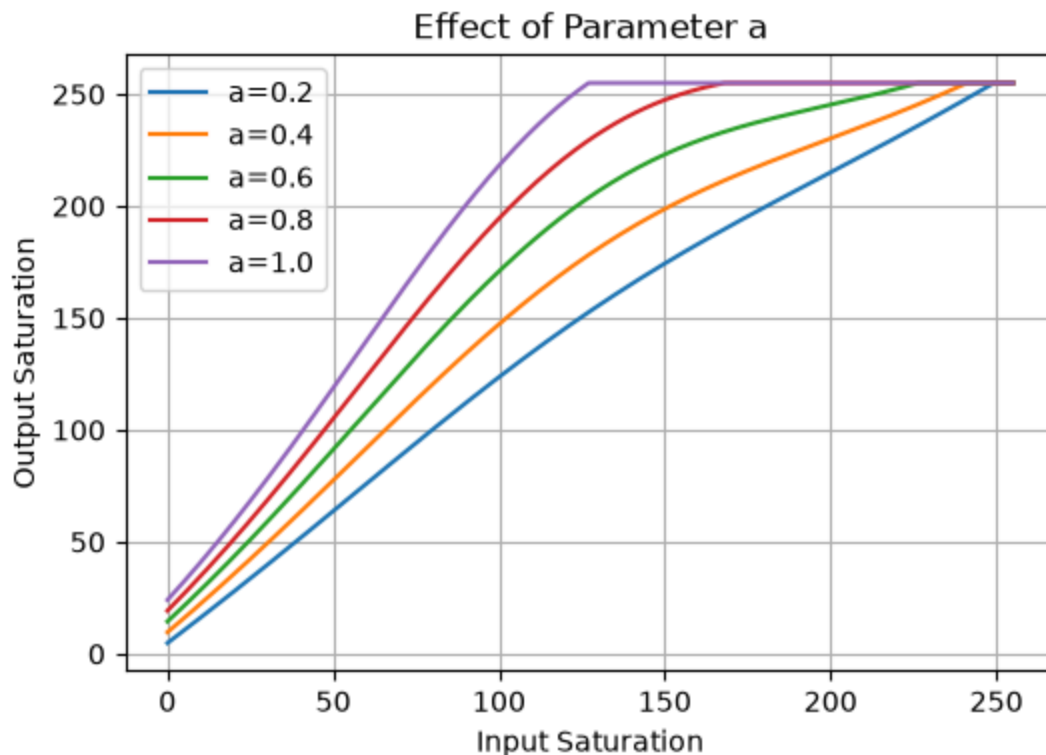


```
In [18]: x = np.arange(256)

plt.figure(figsize=(6,4))

for a_val in a_values:
    y = x + a_val*128*np.exp(-((x-128)**2)/(2*70**2))
    plt.plot(x, np.clip(y,0,255), label=f"a={a_val}")

plt.xlabel("Input Saturation")
plt.ylabel("Output Saturation")
plt.title("Effect of Parameter a")
plt.legend()
plt.grid()
plt.show()
```



Discussion: Increasing (a) increases the saturation boost, with the strongest effect occurring around the middle saturation range. At (a=0.2) and (a=0.4), the change is relatively mild, while (a=0.8) and (a=1.0) produce noticeably stronger colors and some regions approach saturation clipping. A value of (a=0.6) was selected as a compromise because it gives a clear improvement in color vibrance while still keeping the image reasonably natural.

Q5 — Histogram Equalization

Histogram equalization was implemented using the image histogram and its cumulative distribution to spread the grayscale intensities over a wider range.

```
In [17]: # Q5 - Histogram equalization

img_q5 = cv.imread(str(IMAGE_DIR/"q5_histogram.png"), cv.IMREAD_GRAYSCALE)

hist = np.bincount(img_q5.ravel(), minlength=256)
cdf = np.cumsum(hist)
cdf_min = cdf[cdf > 0][0]

mapping = np.clip(
    (cdf - cdf_min) / (img_q5.size - cdf_min) * 255,
    0, 255
).astype(np.uint8)

eq_q5 = mapping[img_q5]

fig,ax = plt.subplots(2,2,figsize=(8,5))
```

```

ax[0,0].imshow(img_q5,cmap="gray",vmin=0,vmax=255)
ax[0,0].set_title("Original")

ax[0,1].imshow(eq_q5,cmap="gray",vmin=0,vmax=255)
ax[0,1].set_title("Equalized")

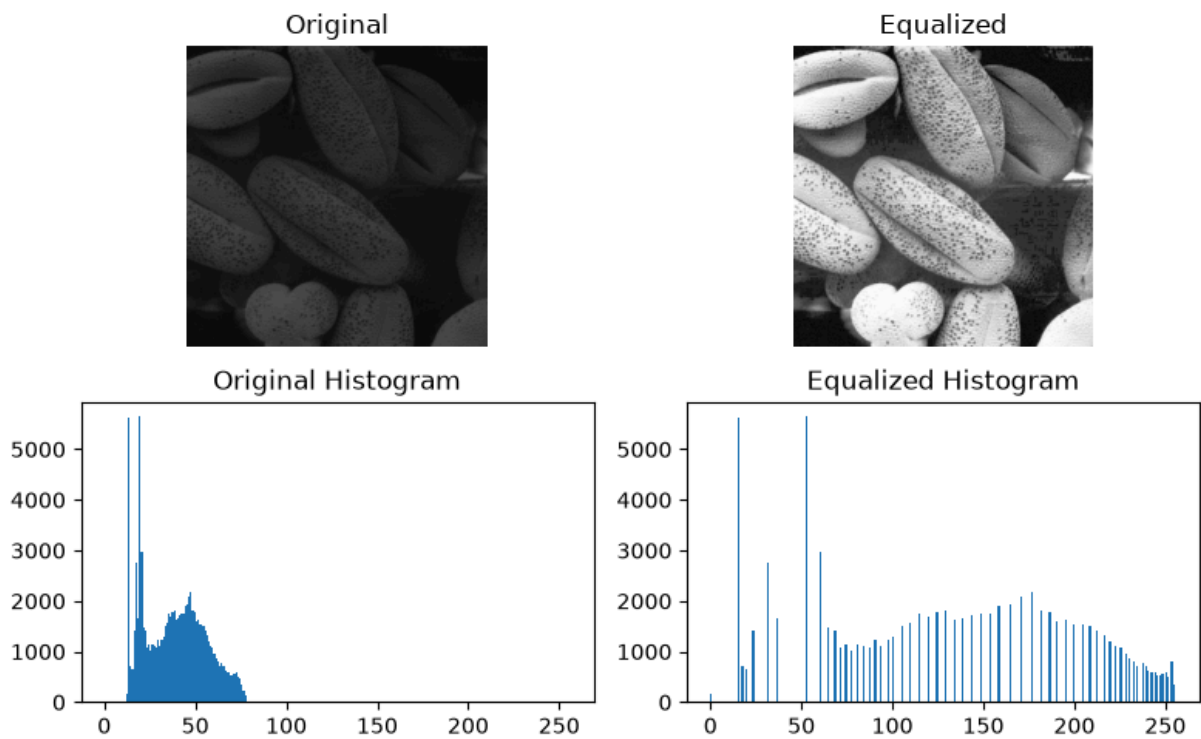
ax[1,0].hist(img_q5.ravel(),bins=256, range=(0,256))
ax[1,0].set_title("Original Histogram")

ax[1,1].hist(eq_q5.ravel(),bins=256, range=(0,256))
ax[1,1].set_title("Equalized Histogram")

for a in ax[0]:
    a.axis("off")

plt.tight_layout()
plt.show()

```



Discussion: The original image uses only a limited part of the grayscale range. Histogram equalization spreads the intensities over a wider range, improving global contrast and making darker details easier to distinguish. The resulting histogram is more widely distributed, although it is not perfectly uniform.

Q6 — Foreground Histogram Equalization

The saturation plane was used to separate the foreground from the gray background. Histogram equalization was then applied only to the foreground pixels of the value channel, while the background was kept unchanged.

```
In [9]: # Q6 - Foreground-only histogram equalization

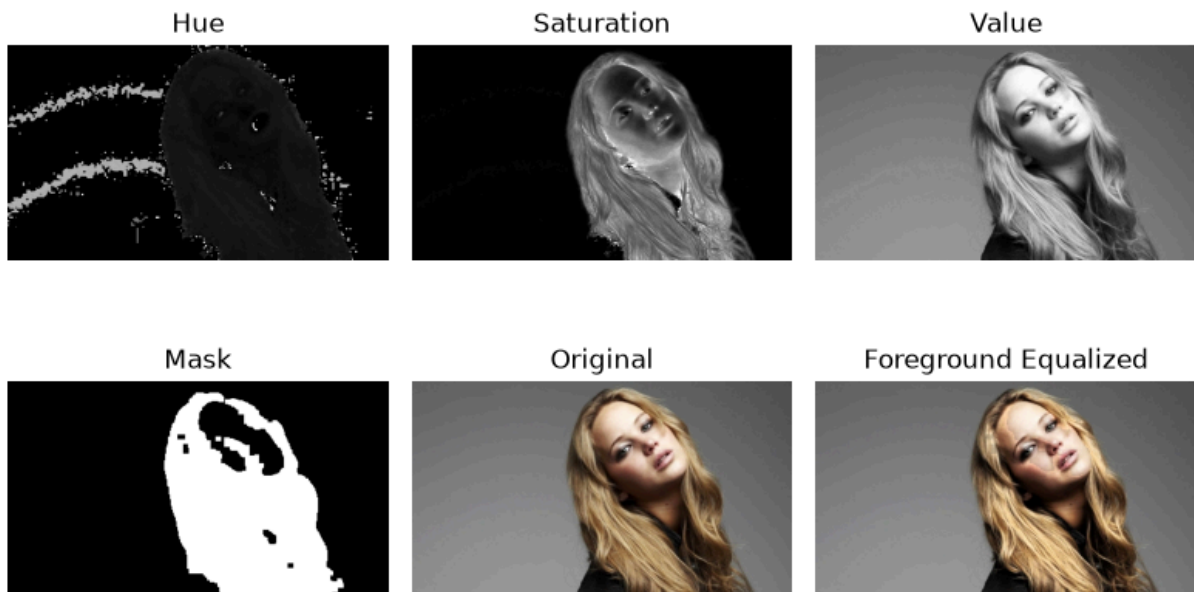
img_q6 = cv.imread(str(IMAGE_DIR/"q6_foreground.png"))
hsv = cv.cvtColor(img_q6, cv.COLOR_BGR2HSV)
H,S,V = cv.split(hsv)

_,mask = cv.threshold(S,80,255,cv.THRESH_BINARY)
k = np.ones((5,5),np.uint8)
mask = cv.morphologyEx(mask,cv.MORPH_CLOSE,k)
mask = cv.morphologyEx(mask,cv.MORPH_OPEN,k)

fg = V[mask==255]
hist = np.bincount(fg,minlength=256)
cdf = np.cumsum(hist); cmin = cdf[cdf>0][0]
lut = np.clip((cdf-cmin)/(fg.size-cmin)*255,0,255).astype(np.uint8)

Veq = V.copy()
Veq[mask==255] = lut[V[mask==255]]
out_q6 = cv.cvtColor(cv.merge([H,S,Veq]),cv.COLOR_HSV2RGB)

fig,ax = plt.subplots(2,3,figsize=(8,5))
for a,im,t in zip(ax.ravel(),[H,S,V,mask,cv.cvtColor(img_q6,cv.COLOR_BGR2RGB),out_q6,
                           ["Hue","Saturation","Value","Mask","Original","Foreground Equaliz
a.imshow(im,cmap="gray" if im.ndim==2 else None); a.set_title(t); a.axis("off")
plt.tight_layout(); plt.show()
```



Discussion: The saturation plane gave a clear separation between the subject and the gray background. Histogram equalization was calculated only from the foreground pixels of the value channel, so the subject contrast was improved while the background remained mostly unchanged.

Q7 — Sobel Filtering

Sobel filtering was carried out using OpenCV `filter2D`, a manual neighborhood implementation, and the separable form of the Sobel kernel.

In [10]: *# Q7 - Sobel filtering using three methods*

```
img_q7 = cv.imread(str(IMAGE_DIR/"q7_sobel.png"), cv.IMREAD_GRAYSCALE)
kx = np.array([[ -1, 0, 1], [ -2, 0, 2], [ -1, 0, 1]], np.float32)
ky = kx.T

# 1) OpenCV filter2D
gx1 = cv.filter2D(img_q7, cv.CV_32F, kx)
gy1 = cv.filter2D(img_q7, cv.CV_32F, ky)
mag1 = np.sqrt(gx1**2+gy1**2)

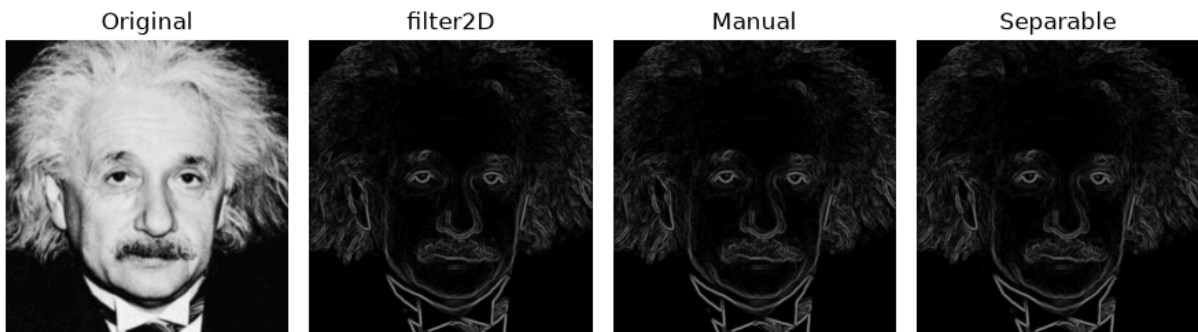
# 2) Manual filtering
def my_filter(im,k):
    p=cv.copyMakeBorder(im.astype(np.float32),1,1,1,1,cv.BORDER_REFLECT)
    out=np.zeros_like(im,dtype=np.float32)
    for y in range(im.shape[0]):
        for x in range(im.shape[1]):
            out[y,x]=np.sum(p[y:y+3,x:x+3]*k)
    return out

gx2,gy2 = my_filter(img_q7,kx),my_filter(img_q7,ky)
mag2 = np.sqrt(gx2**2+gy2**2)

# 3) Separable Sobel
s=np.array([[1],[2],[1]], np.float32); d=np.array([[ -1, 0, 1]], np.float32)
gx3=cv.filter2D(cv.filter2D(img_q7.astype(np.float32), cv.CV_32F, s), cv.CV_32F, d)
gy3=cv.filter2D(cv.filter2D(img_q7.astype(np.float32), cv.CV_32F, d.T), cv.CV_32F, s.T)
mag3=np.sqrt(gx3**2+gy3**2)

fig,ax=plt.subplots(1,4,figsize=(9,3))
for a,im,t in zip(ax,[img_q7,mag1,mag2,mag3],
                  ["Original","filter2D","Manual","Separable"]):
    a.imshow(im,cmap="gray"); a.set_title(t); a.axis("off")
plt.tight_layout(); plt.show()

print("MSE manual:",np.mean((mag1-mag2)**2))
print("MSE separable:",np.mean((mag1-mag3)**2))
```



MSE manual: 1856.9062

MSE separable: 0.0

Discussion: All three methods produce similar Sobel edge responses. The manual implementation performs the neighborhood multiplication directly, while the separable method applies two 1D filters instead of one 2D filter, reducing the number of operations.

Q8 — Image Zooming

Nearest-neighbor and bilinear interpolation were implemented for image zooming. The supplied small images were enlarged by a factor of four and compared with their corresponding original images using normalized SSD.

```
In [11]: # Q8 - Image zooming

def zoom_image(im,s,method):
    h,w=im.shape[:2]
    out=np.zeros((int(h*s),int(w*s),3),dtype=np.uint8)
    for yo in range(out.shape[0]):
        for xo in range(out.shape[1]):
            y,x=yo/s,xo/s
            if method=="nearest":
                yi,xi=min(round(y),h-1),min(round(x),w-1)
                out[yo,xo]=im[yi,xi]
            else:
                y0,x0=int(np.floor(y)),int(np.floor(x))
                y1,x1=min(y0+1,h-1),min(x0+1,w-1)
                dy,dx=y-y0,x-x0
                p00,p10=im[y0,x0].astype(float),im[y0,x1].astype(float)
                p01,p11=im[y1,x0].astype(float),im[y1,x1].astype(float)
                out[yo,xo]=np.clip((1-dx)*(1-dy)*p00+dx*(1-dy)*p10+
                                   (1-dx)*dy*p01+dx*dy*p11,0,255)

    return out

def nssd(ref,test):
    h,w=min(ref.shape[0],test.shape[0]),min(ref.shape[1],test.shape[1])
    ref,test=ref[:h,:w].astype(float),test[:h,:w].astype(float)
    return np.sum((ref-test)**2)/np.sum(ref**2)

pairs=[("im01.png","im01small.png"),("im02.png","im02small.png"),
        ("im03.png","im03small.png"),("im04.jpg","im04small.jpg")]

for a,b in pairs:
    ref,small=cv.imread(str(IMAGE_DIR/a)),cv.imread(str(IMAGE_DIR/b))
    nn,bi=zoom_image(small,4,"nearest"),zoom_image(small,4,"bilinear")
    print(f"{a}: Nearest={nssd(ref,nn):.5f}, Bilinear={nssd(ref,bi):.5f}")
```

```
im01.png: Nearest=0.02259, Bilinear=0.01772
im02.png: Nearest=0.01024, Bilinear=0.00776
im03.png: Nearest=0.01472, Bilinear=0.01113
im04.jpg: Nearest=0.00960, Bilinear=0.00771
```

```
In [12]: ref=cv.imread(str(IMAGE_DIR/"im01.png"))
small=cv.imread(str(IMAGE_DIR/"im01small.png"))
nn,bi=zoom_image(small,4,"nearest"),zoom_image(small,4,"bilinear")

fig,ax=plt.subplots(1,3,figsize=(8,3))
for a,im,t in zip(ax,[ref,nn,bi],["Original","Nearest","Bilinear"]):
    a.imshow(cv.cvtColor(im,cv.COLOR_BGR2RGB)); a.set_title(t); a.axis("off")
plt.tight_layout(); plt.show()
```



Discussion: Nearest-neighbor interpolation gives a more block-like result, while bilinear interpolation produces smoother transitions by using four surrounding pixels. The lower NSSD values indicate which interpolation method reconstructs the supplied original images more closely.

Q9 — Foreground Segmentation and Background Blur

GrabCut was used to separate the main flower from the background. The extracted background was blurred and then recombined with the sharp foreground.

```
In [13]: # Q9 - GrabCut segmentation and background blur

img_q9=cv.imread(str(IMAGE_DIR/"q9_flower.png"))
h,w=img_q9.shape[:2]

mask=np.zeros((h,w),np.uint8)
bg=np.zeros((1,65),np.float64); fg=np.zeros((1,65),np.float64)
rect=(int(.1*w),int(.05*h),int(.8*w),int(.85*h))

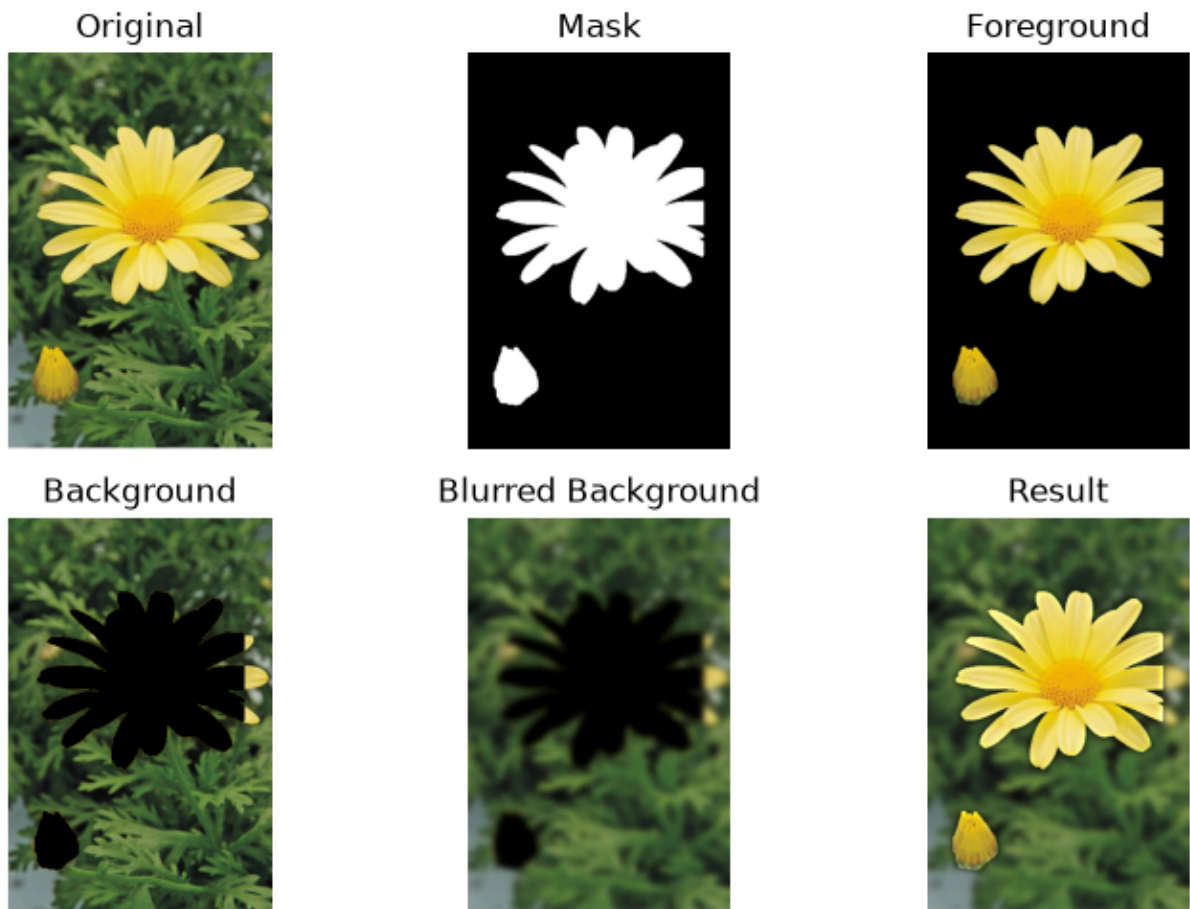
cv.grabCut(img_q9,mask,rect,bg,fg,5,cv.GC_INIT_WITH_RECT)

m=np.where((mask==cv.GC_FGD)|(mask==cv.GC_PR_FGD),255,0).astype(np.uint8)
fg_img=cv.bitwise_and(img_q9,img_q9,mask=m)
bg_mask=cv.bitwise_not(m)
bg_img=cv.bitwise_and(img_q9,img_q9,mask=bg_mask)

blur_bg=cv.GaussianBlur(bg_img,(31,31),0)
out_q9=cv.add(fg_img,blur_bg)

imgs=[img_q9,m,fg_img,bg_img,blur_bg,out_q9]
titles=["Original","Mask","Foreground","Background","Blurred Background","Result"]

fig,ax=plt.subplots(2,3,figsize=(8,5))
for a,im,t in zip(ax.ravel(),imgs,titles):
    if im.ndim==2: a.imshow(im,cmap="gray")
    else: a.imshow(cv.cvtColor(im,cv.COLOR_BGR2RGB))
    a.set_title(t); a.axis("off")
plt.tight_layout(); plt.show()
```



Discussion: GrabCut successfully separated the main flower from the background. Blurring the extracted background makes the foreground more prominent. A dark halo can appear near the flower boundary because the removed foreground area contains zero-valued pixels, which are included in the Gaussian averaging near the edge.

Q10 — Bilateral Filtering

Gaussian and bilateral filtering were compared. A custom bilateral filter was also implemented and compared with OpenCV's `bilateralFilter`.

```
In [14]: img_q10 = cv.imread(str(IMAGE_DIR/"q10_lake.png"))
ks, ss, sr = 5, 3, 30

gauss = cv.GaussianBlur(img_q10,(ks,ks),ss)
bil_cv = cv.bilateralFilter(img_q10,ks,sr,ss)

def bilateral(im,ks,ss,sr):
    r=ks//2; p=cv.copyMakeBorder(im.astype(float),r,r,r,r,cv.BORDER_REFLECT); out=np
    Y,X=np.mgrid[-r:r+1,-r:r+1]; ws=np.exp(-(X**2+Y**2)/(2*ss**2))
    for y in range(im.shape[0]):
        for x in range(im.shape[1]):
            R=p[y:y+ks,x:x+ks]; c=p[y+r,x+r]
            wr=np.exp(-np.sum((R-c)**2,axis=2)/(2*sr**2)); w=ws*wr
            out[y,x]=np.sum(R*w[:, :, None],axis=(0,1))/np.sum(w)
    return np.clip(out,0,255).astype(np.uint8)
```

```

bil_my = bilateral(img_q10,ks,ss,sr)
mse = np.mean((bil_cv.astype(float)-bil_my.astype(float))*2)
psnr = 10*np.log10(255**2/mse)
print(f"MSE = {mse:.3f}, PSNR = {psnr:.2f} dB")

```

MSE = 5.900, PSNR = 40.42 dB

```

In [15]: imgs=[img_q10,gauss,bil_cv,bil_my]
titles=["Original","Gaussian","OpenCV Bilateral","Custom Bilateral"]

plt.figure(figsize=(12,3))
for i,(im,t) in enumerate(zip(imgs,titles)):
    plt.subplot(1,4,i+1); plt.imshow(cv.cvtColor(im,cv.COLOR_BGR2RGB))
    plt.title(t); plt.axis("off")
plt.tight_layout(); plt.show()

```



Discussion

Gaussian filtering smooths using only spatial distance, so edges are also blurred. Bilateral filtering additionally considers intensity/color difference, allowing smoothing while preserving strong edges. The custom result was compared with OpenCV using MSE and PSNR; lower MSE and higher PSNR indicate closer agreement.